

cxreg: An R package for complex-valued lasso and graphical lasso

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Summary

Pathwise coordinate descent (Friedman et al., 2007) is a widely used iterative optimization method for solving penalized linear regression (*lasso*; (Tibshirani, 1996)) and penalized Gaussian log-likelihood (*graphical lasso*; e.g., Banerjee, El Ghaoui, and d'Aspremont (Banerjee et al., 2008)) problems. For a given sequence of tuning parameters λ in the penalty term, the algorithm updates one coefficient at a time by computing partial residuals.

The coordinate descent algorithm for complex-valued lasso and Gaussian graphical lasso, implemented in the *cxreg* R package, extends the standard pathwise coordinate descent method to complex-valued data (Deb, Kuceyeski, and Basu (Deb et al., 2024)). The algorithm operates by *realifying* complex numbers through a ring isomorphism (Herstein, 1991). Specifically, for a complex number $z \in \mathbb{C}$, we define the map:

$$\varphi(z) = \begin{pmatrix} \text{Re}(z) & -\text{Im}(z) \\ \text{Im}(z) & \text{Re}(z) \end{pmatrix}.$$

It is known that φ is a field isomorphism between $\mathbb{C}$ and the set $\mathcal{M}^{2 \times 2}$ whose entries consists of $a, b \in \mathbb{R}$ to form 2 by 2 matrices (Artin, 2018),

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Under this mapping, algebraic operations in the complex domain correspond directly to those in the real domain. We extend these operations to the p -dimensional case ($p > 1$); see Section 3.1 of Deb, Kuceyeski, and Basu (Deb et al., 2024) for details.

Using this correspondence, we transform complex-valued linear regression problems into equivalent real-valued formulations. The same idea applies to the Gaussian log-likelihood for complex-valued variables. For the ℓ_1 penalty used in lasso and graphical lasso, we note that the modulus of a complex coefficient equals the entrywise ℓ_2 norm of its real and imaginary parts. Consequently, the lasso for p -dimensional complex variables is equivalent to a $2p$ -dimensional group lasso (Yuan and Lin (Yuan & Lin, 2006)), where each group contains the real and imaginary components of one complex variable.

We also incorporate standard computational enhancements, including warm starts, using solutions from previous λ values, and active set selection to limit updates to relevant predictors (see Hastie, Tibshirani, and Wainwright (Hastie et al., 2015), Chapter 5.4). The R package *glmnet* adopts a familiar interface similar to *glmnet* (Friedman, Hastie, and Tibshirani (Friedman et al., 2010)), and a detailed vignette is available [Here](#).

Statement of need

Solving complex-valued penalized Gaussian likelihood problems (and, as a component, penalized linear regression) is central to high-dimensional time series analysis. In the frequency domain, examining partial spectral coherence is analogous to studying partial correlations in Gaussian graphical models (e.g., Priestley ([Priestley, 1988](#))). With the development of the local Whittle likelihood approximation (Whittle ([Whittle, 1951](#))), the inverse spectral density matrix, known as the spectral precision matrix, provides a way to characterize associations among variables in high-dimensional time series data.

These associations, represented by the spectral precision matrix, are evaluated at fixed frequencies. However, because the spectral density matrix estimator is inconsistent at a single frequency (see Brockwell and Davis ([Brockwell & Davis, 1991](#)), Chapter 10), it is common to use an averaged, smoothed periodogram, obtained by aggregating periodogram matrices across neighboring frequencies, as a consistent sample analog. Consequently, spectral precision matrices must be computed across multiple neighboring frequencies around each frequency of interest. Such analyses involve repeated high-dimensional optimization, as required for statistical inference on spectral precision matrices (Krampe and Paparoditis ([Krampe & Paparoditis, 2025](#))). Therefore, developing efficient algorithms for complex-valued penalized Gaussian likelihood estimation is crucial for scalable and accurate frequency-domain analysis.

State of the field

Although coordinate descent is computationally efficient and widely used to solve the lasso (Friedman et al. ([Friedman et al., 2007](#))) and graphical lasso (Friedman, Hastie, and Tibshirani ([Friedman et al., 2008](#))), particularly under sparsity assumptions, its extension to complex-valued settings remains limited. This gap largely arises from the lack of methods for updating complex-valued partial residuals within the coordinate descent framework.

Several alternative approaches have been proposed for complex-valued graphical modeling, but many do not explicitly exploit sparsity within a coordinate descent framework, which can limit scalability in high-dimensional settings. Recent software, such as the R package *tsglasso* (Dallakyan et al. ([Dallakyan et al., 2022](#))) and the Python package *FreDom* (Dallakyan ([Dallakyan, 2024](#))), provide related implementations based on closed-form updates derived via Wirtinger calculus. Another line of work employs the alternating direction method of multipliers (ADMM; Baek, Düker, and Pipiras ([Baek et al., 2023](#))).

In contrast, the algorithm implemented in this package is specifically designed to leverage sparsity within a coordinate descent framework. The method follows ([Deb et al., 2024](#)), where comparisons with benchmark approaches, including the complex-valued extension of the nodewise regression CIPE algorithm (Fiecas et al. ([Fiecas et al., 2019](#))) and its pathwise variants, demonstrate improvements in both estimation accuracy and computational efficiency.

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Artin, M. (2018). *Algebra*. Pearson.

Baek, C., Düker, M.-C., & Pipiras, V. (2023). Local Whittle estimation of high-dimensional long-run variance and precision matrices. *The Annals of Statistics*, 51(6), 2386–2414.

Banerjee, O., El Ghaoui, L., & d'Aspremont, A. (2008). Model selection through sparse

- maximum likelihood estimation for multivariate gaussian or binary data. *The Journal of Machine Learning Research*, 9, 485–516.
- Brockwell, P. J., & Davis, R. A. (1991). *Time series: Theory and methods*. Springer science & business media.
- Dallakyan, A. (2024). On learning time series DAGs: A frequency domain approach. *Econometrics and Statistics*.
- Dallakyan, A., Kim, R., & Pourahmadi, M. (2022). Time series graphical lasso and sparse VAR estimation. *Computational Statistics & Data Analysis*, 176, 107557.
- Deb, N., Kuceyeski, A., & Basu, S. (2024). Regularized estimation of sparse spectral precision matrices. *arXiv Preprint arXiv:2401.11128*.
- Fiecas, M., Leng, C., Liu, W., & Yu, Y. (2019). Spectral analysis of high-dimensional time series. *Electronic Journal of Statistics*, 13(2), 4079–4101.
- Friedman, J., Hastie, T., Höfling, H., & Tibshirani, R. (2007). Pathwise coordinate optimization. *The Annals of Applied Statistics*, 302–332.
- Friedman, J., Hastie, T., & Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3), 432–441.
- Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33, 1–22.
- Hastie, T., Tibshirani, R., & Wainwright, M. (2015). Statistical learning with sparsity. *Monographs on Statistics and Applied Probability*, 143(143), 8.
- Herstein, I. N. (1991). *Topics in algebra*. John Wiley & Sons.
- Krampe, J., & Paparoditis, E. (2025). Frequency domain statistical inference for high-dimensional time series. *Journal of the American Statistical Association*, 1–13.
- Priestley, M. B. (1988). *The spectral analysis of time series*. Oxford University Press.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1), 267–288.
- Whittle, P. (1951). Hypothesis testing in time series analysis. (*No Title*).
- Yuan, M., & Lin, Y. (2006). Model selection and estimation in regression with grouped variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 68(1), 49–67.